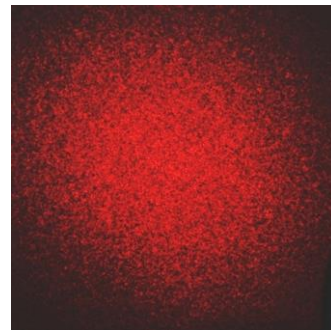
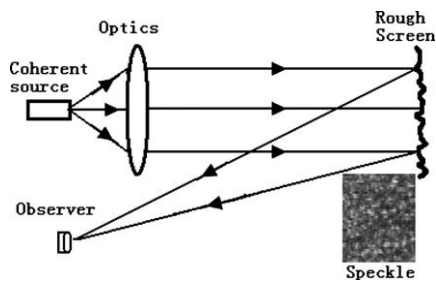


# Laser speckle contrast imaging and multiple scattering theory

## Speckle

For coherent illumination sources, light scattering generates a speckle pattern



## Basic equations of speckle

- ▶ A rough surface with random scattering positions and amplitudes,  
 $A(x, y) = N^{-1/2} \sum_{m=1}^n |A_m| \exp(-i\phi_m)$
- ▶ Uniformly distributed phase in  $[0, 2\pi]$  and Gaussian distribution of real and imaginary parts
- ▶ Probability density function for the intensity:  $P(I) = \frac{1}{\sigma^2} \exp\left(-\frac{I}{\sigma^2}\right)$
- ▶ Speckle size as seen via a lens with a diameter  $D$  and focal length  $f$ :  
 $0.61\lambda f/D$

## Speckle dynamics

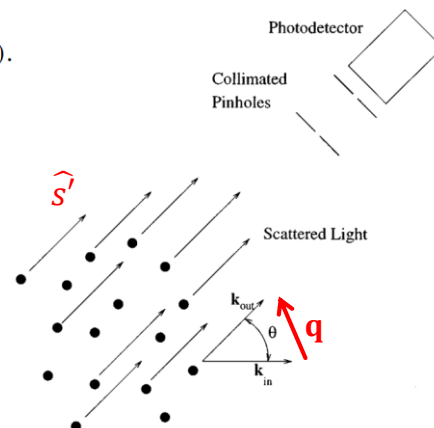
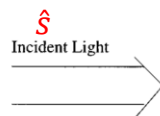
$$\mathbf{E} = \mathbf{E}_0 F(q) \sum_{n=1}^N \exp(i\mathbf{k}_{\text{in}} \cdot \mathbf{r}_n) \exp[i\mathbf{k}_{\text{out}} \cdot (\mathbf{R}_d - \mathbf{r}_n)]$$

$$= \mathbf{E}_0 F(q) \exp(i\mathbf{k}_{\text{out}} \cdot \mathbf{R}_d) \sum_{n=1}^N \exp(-i\mathbf{q} \cdot \mathbf{r}_n).$$

For moving particles:

$$g_1(\mathbf{q}, \tau) = \frac{\langle E(0)E^*(\tau) \rangle}{\langle |E(0)|^2 \rangle} =$$

$$\int w(\mathbf{r}, \tau) \exp(-i\mathbf{q} \cdot \mathbf{r})$$

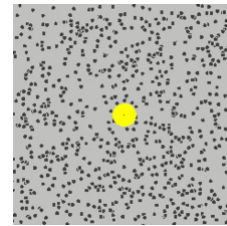


D. Boas and A. G. Yodh, J. Opt. Soc. Am. A **14**, 1997

## Brownian motion

- The probability density function:

$$w(\mathbf{r}, \tau) = \exp \left[ \frac{-|\mathbf{r}|^2}{4D_0\tau} \right] / (4\pi D_0\tau)^{3/2}$$



- The temporal field correlation function

$$g_1(\mathbf{q}, \tau) = \exp(-D_0|\mathbf{q}|^2 \cdot \tau) = \exp[-2D_0k_{in}^2\tau(1 - \hat{s} \cdot \hat{s}')] ]$$

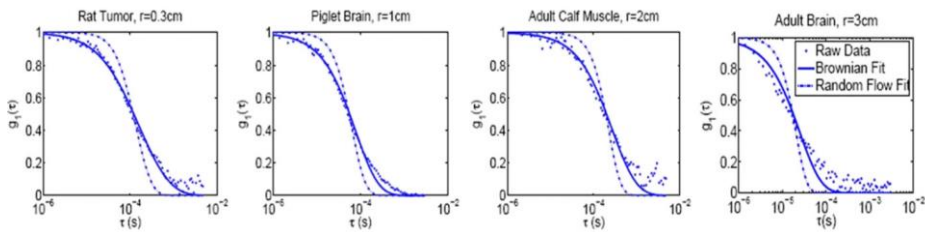
## Siegert relation

- Measuring correlations of fields,  $G_1(\tau) = \langle E(0)E^*(\tau) \rangle$ , is difficult.
- Measuring intensities  $I(t) \propto \langle |E(t)|^2 \rangle$  is easy.
- Defining  $G_2(\tau) = \langle I(0)I(\tau) \rangle$ , Siegert relation is

$$G_2(\tau) = \langle I \rangle^2 + \beta |G_1(\tau)|^2$$

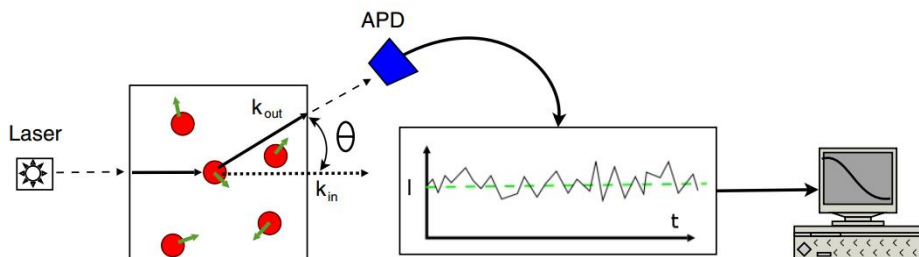
## Statistics of blood flow

Blood flow in the body is not Brownian, but the complexity of blood flow and vasculature leads to the  $g_1(\mathbf{q}, \tau)$  obtained by the Brownian model

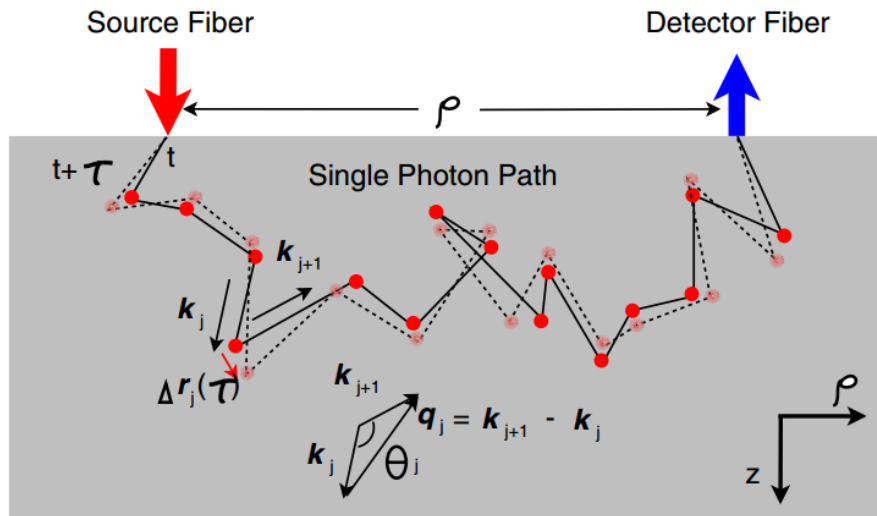


T. Durduran *et al.*, Rep. Prog. Phys. **73**, 2010

## Single scattering limit



## Multiple scattering



## RTE for the correlation function

$$\hat{\mathbf{s}} \cdot [\nabla G(\mathbf{r}, \hat{\mathbf{s}}, \tau)] + \mu_t G(\mathbf{r}, \hat{\mathbf{s}}, \tau)(\mathbf{r}, \hat{\mathbf{s}}) - \mu_s \int_{4\pi} G(\mathbf{r}, \hat{\mathbf{s}}', \tau) |f(\hat{\mathbf{s}}', \hat{\mathbf{s}})|^2 g_1(K_r(\hat{\mathbf{s}} - \hat{\mathbf{s}}'), \tau) d\Omega' = \epsilon(r, \hat{\mathbf{s}})$$

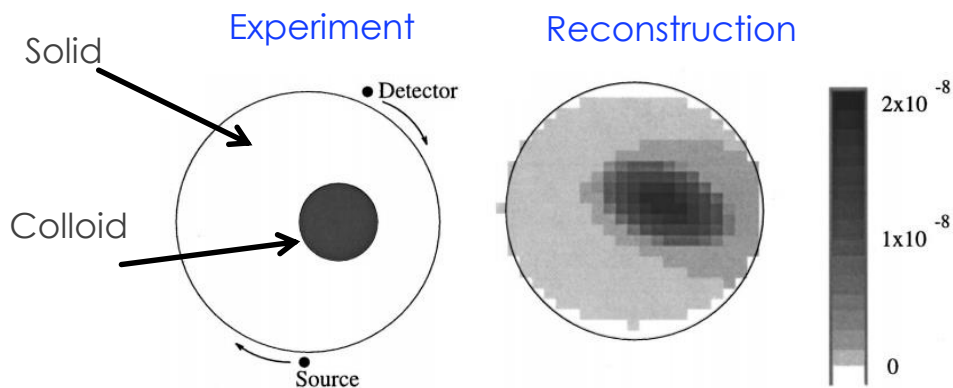
## Correlation diffusion equation

$$\left[ D \nabla^2 - \mu_a - \frac{1}{3} c k_{in}^2 \mu'_s \right] \langle \Delta r^2(\tau) \rangle G(\mathbf{r}, \hat{\mathbf{s}}, \tau) = -S_0(\mathbf{r}, t)$$

For Brownian motion:  $\langle \Delta r^2(\tau) \rangle = 6D_0\tau$

D. Boas and A. G. Yodh, J. Opt. Soc. Am. A **14**, 1997

## Tomographic imaging



D. Boas and A. G. Yodh, J. Opt. Soc. Am. A **14**, 1997

## Laser speckle contrast imaging

- ▶ In time-resolved measurements, we have only few detectors -> long measurements or low resolution
- ▶ Cameras can capture high-resolution image at high frame rate, but involve temporal averaging
- ▶ How do the intensity fluctuations affect the measured image?

## Speckle contrast

For static speckle:

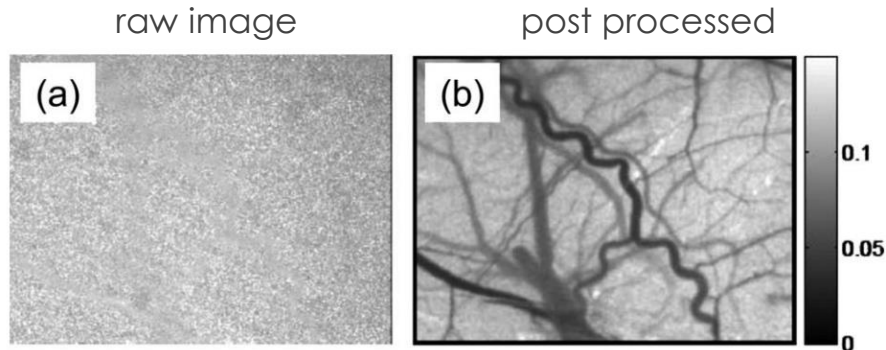
$$P(I) = \frac{1}{\sigma^2} \exp\left(-\frac{I}{\sigma^2}\right)$$

$$E[I] = \sqrt{\text{Var}[I]}$$

Speckle contrast (via spatial averaging):  $K_s = \frac{\sigma_s}{\langle I \rangle}$

For static speckle:  $K_s = 1$ , but if the speckle fluctuates, the temporal averaging of the camera will reduce from  $K_s = 1$

## Image post processing procedure



$K_s$  is calculated over small (7x7) windows

The post-processed image is of  $K_s$

## Speckle contrast for Brownian motion

$$K(t) \cong \frac{1}{T \langle I \rangle^2} \int_0^T C_t(\tau) d\tau$$

$$C_t(\tau) = \langle [I(t) - \langle I \rangle_t][I(t + \tau) - \langle I \rangle_t] \rangle_t$$

$$g_1(\tau) = \exp\left(-\frac{\tau}{\tau_c}\right)$$

$$K(T) = \frac{\sigma_s}{\langle I \rangle} = \left\{ \frac{\tau_c}{2T} \left[ 1 - \exp\left(-\frac{2T}{\tau_c}\right) \right] \right\}^{1/2}$$

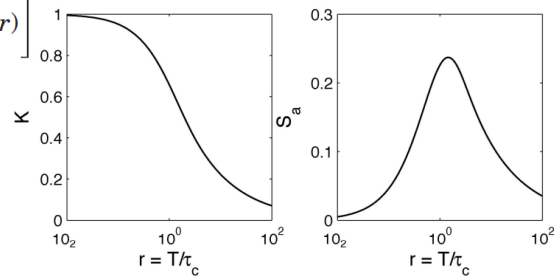
S. Yuan, et al. Appl. Opt. **44**, 2005



## Sensitivity

$$S_a = \left| \frac{dK}{dv} \right| \propto -\tau_c r \frac{dK}{d\left(\frac{T}{\tau_c}\right)} = -\tau_c r \frac{dK}{dr}$$

$$= \tau_c \frac{r}{2K} \left[ \frac{1}{2r^2} - \frac{(2r+1)}{2r^2} \exp(-2r) \right]$$

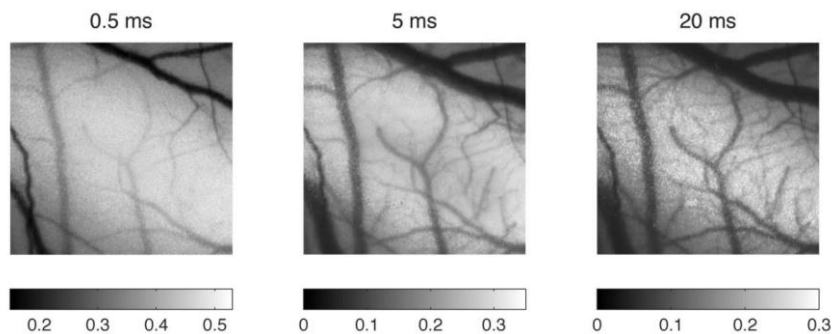


For  $T > \tau_c$  the benefit of larger integration times is diminished

S. Yuan, et al. Appl. Opt. **44**, 2005

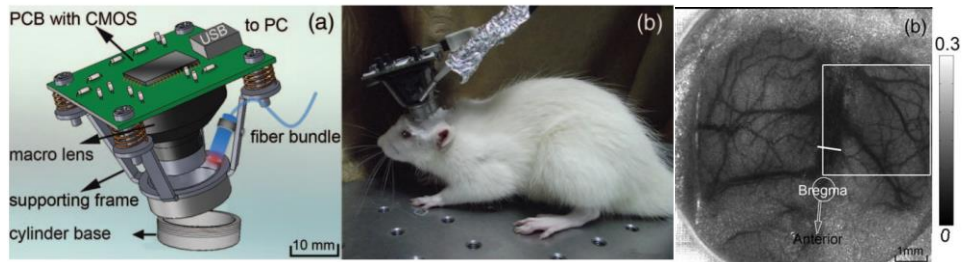
## Heterogeneity in blood flow

Smaller blood vessels have lower flow velocity (small  $\tau_c$ ) and required longer exposure times



S. Yuan, et al. Appl. Opt. **44**, 2005

## Application: monitoring cerebral dynamics in mice



P. Miao et al. JBO Lett. **16** (2011)